Mini Project

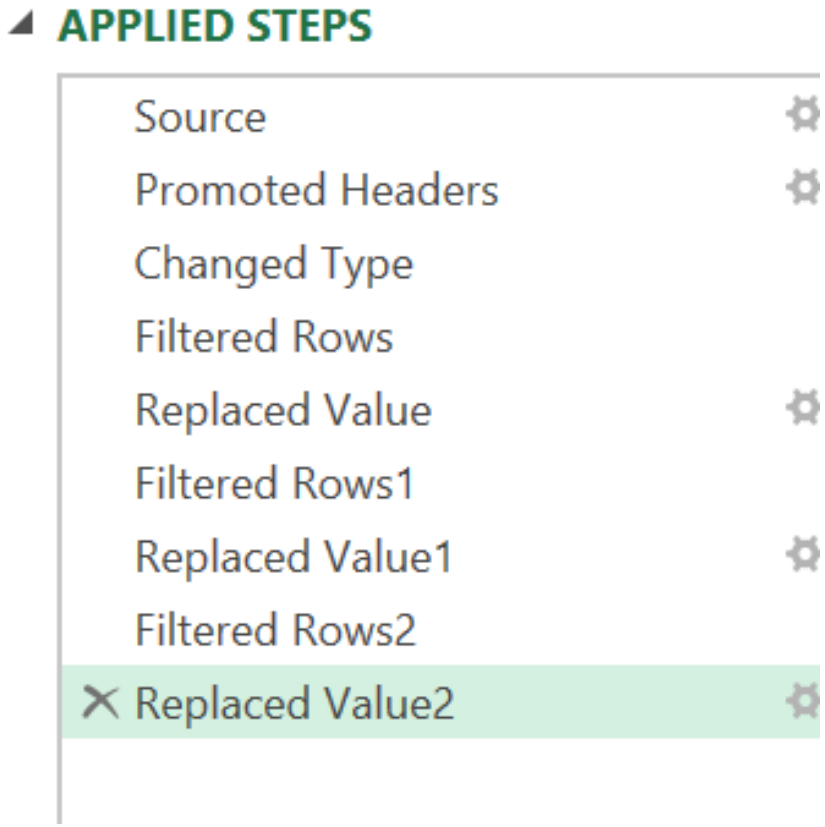
Excel and Power BI

Original Dataset Source

- https://www.kaggle.com/datasets/lainguyn123/student-performance-factors

Cleaned Dataset

- Performed data cleaning and transformation using Power Query
- Identified missing, incorrect, and inconsistent values in the dataset.
- Applied filters to locate rows containing errors and blank entries. Corrected data entry errors by replacing inaccurate values with appropriate values.
- Replaced missing values using suitable methods
- Standardized categorical data to ensure consistency in naming and formatting.
- Removed duplicate records to avoid redundancy.
- Verified data types and formats for each column.
- Conducted a final review to ensure the dataset was clean, complete, and ready for analysis.



Cleaned Dataset

- Conducted descriptive statistics on the cleaned dataset.
- Calculated measures including mean, median, minimum, and maximum values.
- Created Pivot Tables to summarize and analyze data.
- Identified trends, patterns, and relationships among variables.
- Prepared summarized data for visualization and reporting.

Parental Involvement

Row Labels	Average of Exam_Score
High	68.0927673
Low	66.35826477
Medium	67.09815586
Grand Total	**67.23565915**

Motivation level

Row Labels	Average of Tutoring_Sessions
High	1.515542077
Low	1.504904491
Medium	1.478663086
Grand Total	**1.493718783**

Extra curricular activities

Row Labels	Average of Sleep_Hours
No	7.026601723
Yes	7.030726257
Grand Total	**7.029060088**

Column1	
Mean	67.23565915
Standard Error	0.047862825
Median	67
Mode	68
Standard Deviation	3.890455781
Sample Variance	15.13564619
Kurtosis	10.57542273
Skewness	1.644808285
Range	46
Minimum	55
Maximum	101
Sum	444226
Count	6607

Power BI Report

- Imported the cleaned dataset into Power BI.
- Created additional calculated columns, including **Performance Category** and **Hours Studied Category**, to support analysis.
- Developed DAX measures to calculate key performance indicators (KPIs) and summary metrics.
- Designed interactive dashboards and visualizations to explore relationships between variables.
- Added KPI cards, charts, and matrices to present insights effectively.
- Used filters and slicers to enable dynamic data exploration.
- Generated a comprehensive Power BI report to visualize student performance trends and patterns.

Student Performance Dashboard

7K	67.24	79.98	19.98
Total Students	Average Exam Score	Average Attendance	Average Study Hours

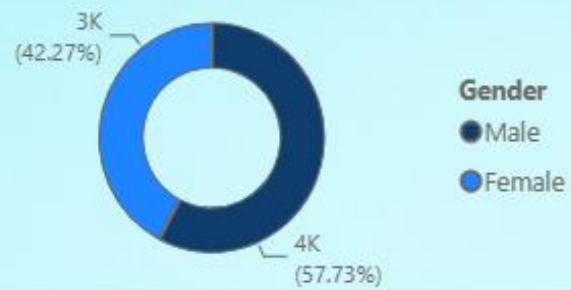
Gender
All

School_Type
All

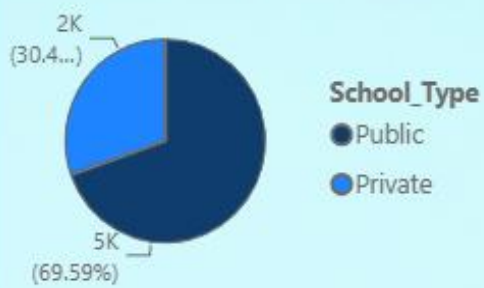
Performance category
All

Family_Income
All

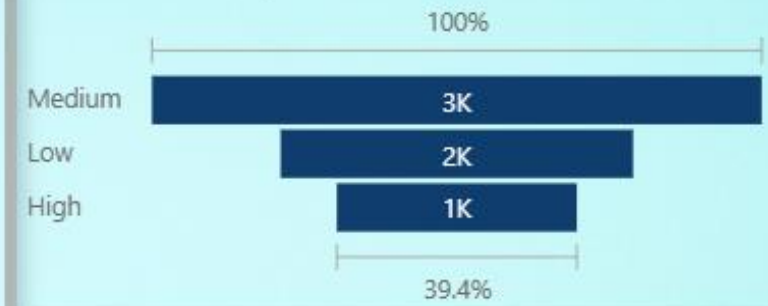
Total Students by Gender



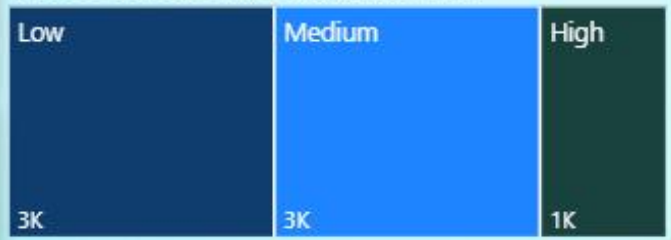
Total Students by School_Type



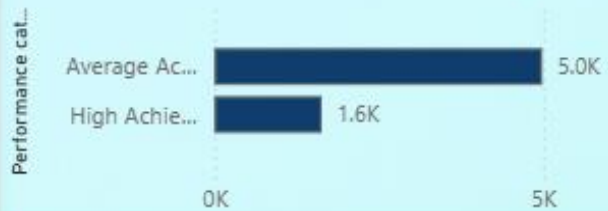
Total Students by Motivation_Level



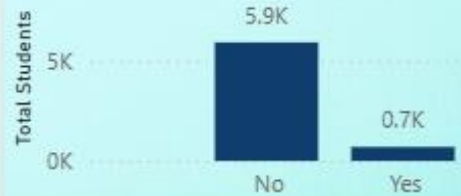
Total Students by Family_Income



Total Students by Performance category



Total Students by Learning_Disabilities



Executive Overview | School Environment & Student Success | Academic Performance | Family & Resource Analysis | Lifestyle & Wellbeing

School Environment & Student Success Analysis

School_Type: All | Peer_Influe...: All | Motivation_L...: All | Teacher_Quality: All

67.21
Average exam score(Public)

67.29
Average exam score(Private)

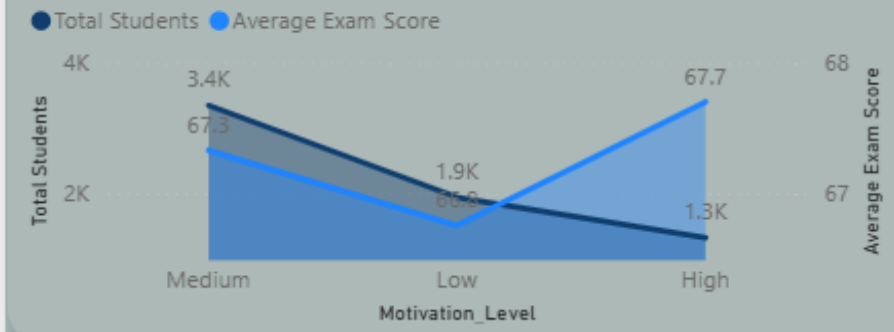
School Performance Comparison by Learning Disability Status

School_Type	Private				Public			
Learning_Disabilities	Total Students	Average Exam Score	Average Attendance	High Achiever%	Total Students	Average Exam Score	Average Attendance	High Achiever%
No	1798	67.41	80.43	26.03	4114	67.32	79.90	25.21
Yes	211	66.23	79.19	16.59	484	66.29	79.24	17.56

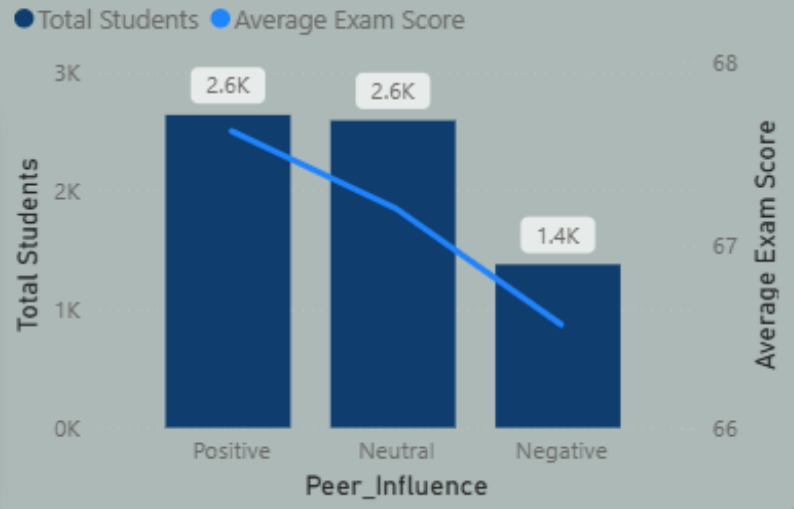
Total Students by School_Type



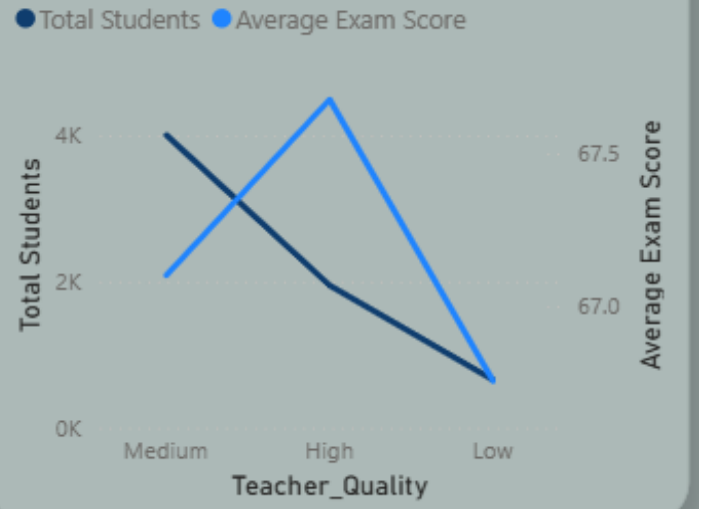
Total Students and Average Exam Score by Motivation_Level



Total Students and Average Exam Score by Peer_Influence



Total Students and Average Exam Score by Teacher_Quality



Executive Overview | School Environment & Student Success | Academic Performance | Family & Resource Analysis | Lifestyle & Wellbeing

Academic Performance Analysis

Gender: All

Motivation_Level: All

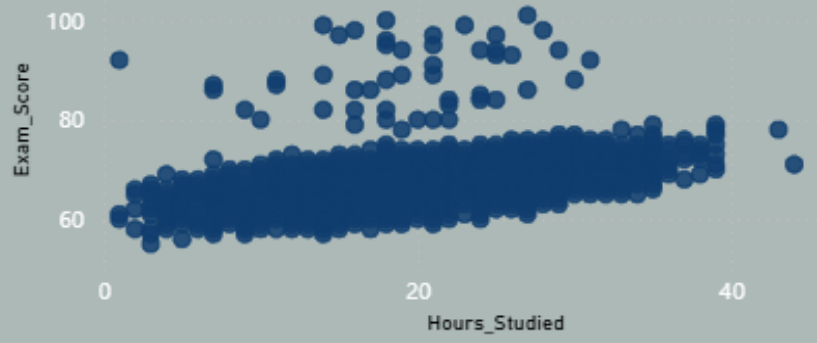
Tutoring_Sessions: 0 – 8

19.98
Average Study Hours

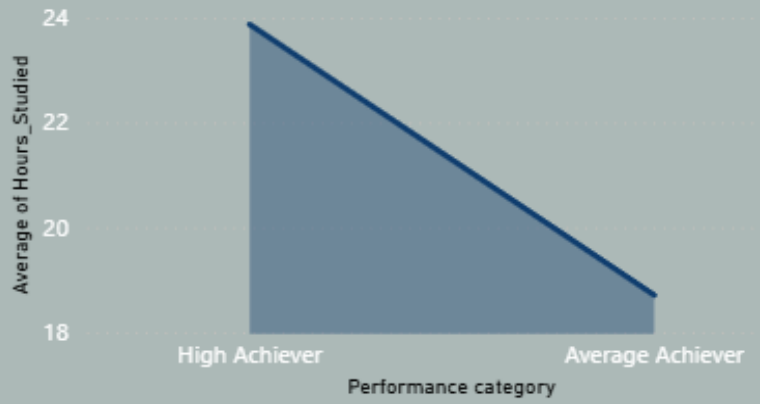
79.98
Average Attendance

75.07
Average of Previous_Scores

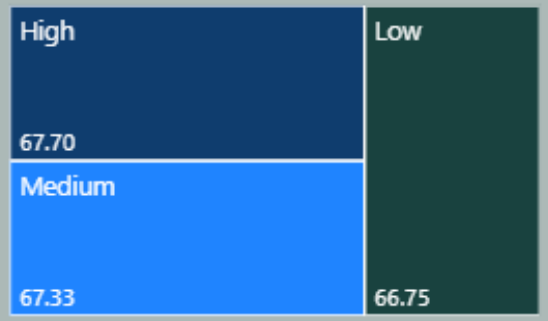
Hours_Studied and Exam_Score



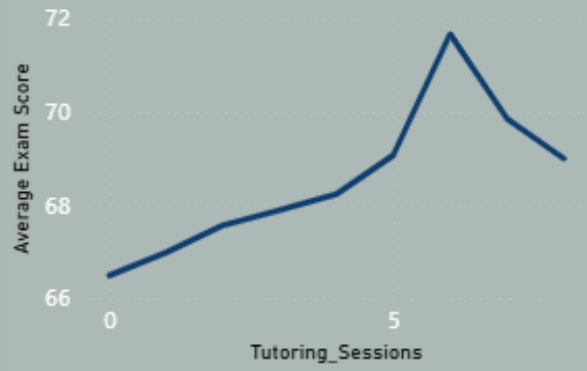
Average of Hours_Studied by Performance category



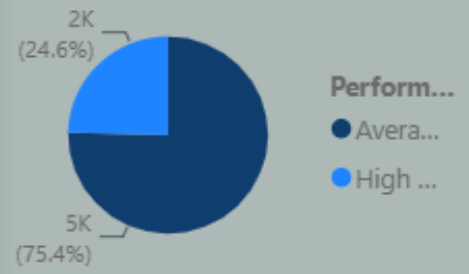
Average Exam Score by Motivation_Level



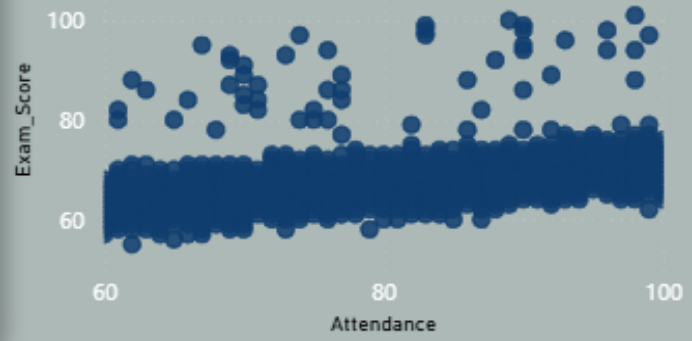
Average Exam Score by Tutoring_Sessions



Total Students by Performance category



Attendance and Exam_Score



Executive Overview | School Environment & Student Success | **Academic Performance** | Family & Resource Analysis | Lifestyle & Wellbeing

Family & Resource Analysis

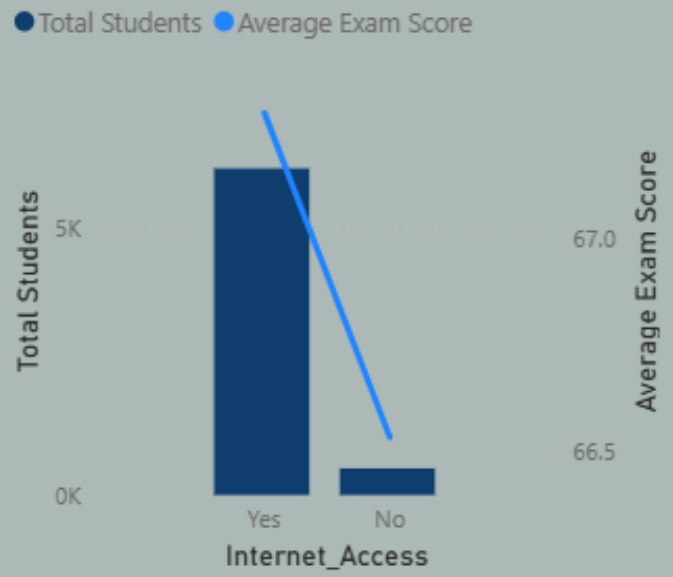
Internet_Ac... All | Access_... All | Family_Income All | Parental_Educat... All

28.88 Students with High Parental involvment%

29.89 Students with High resource access%

92.45 Students with internet access%

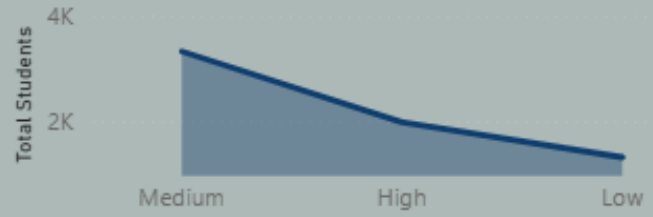
Total Students and Average Exam Score by Internet_Access



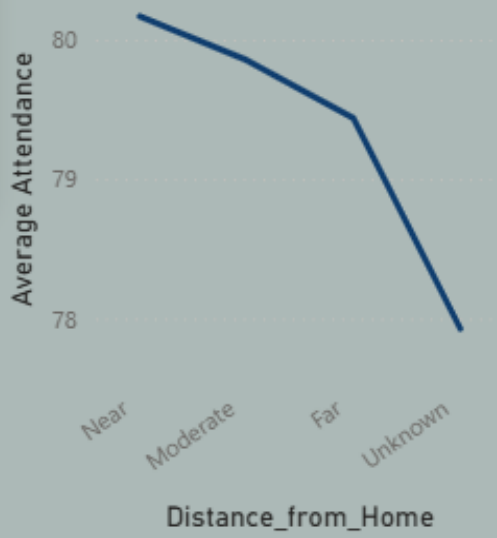
Average Exam score by Parental Education and Involvment

Parental_Education_Level	High	Low	Medium
College	68.22	66.47	67.13
High School	67.78	65.99	66.75
Not specified	68.08	65.85	67.04
Postgraduate	68.68	67.10	67.93

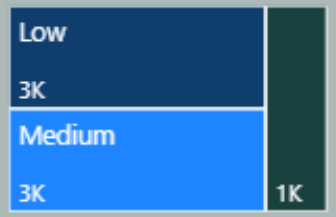
Total Students by Access_to_Resources



Average Attendance by Distance_from_Home



Total Students by Family_Income



Performance Analysis by access to Resources

Performance category	Average Achiever			High Achiever		
Access_to_Resources	Total Students	Average of Hours_Studied	Average of Previous_Scores	Total Students	Average of Hours_Studied	Average of Previous_Scor
High	1300	18.33	73.62	675	23.21	78.7
Low	1112	19.32	73.32	201	24.57	79.9
Medium	2570	18.63	74.22	749	24.29	78.4

Executive Overview | School Environment & Student Success | Academic Performance | Family & Resource Analysis | Lifestyle & Wellbeing

Lifestyle & Wellbeing analysis

Gender: All

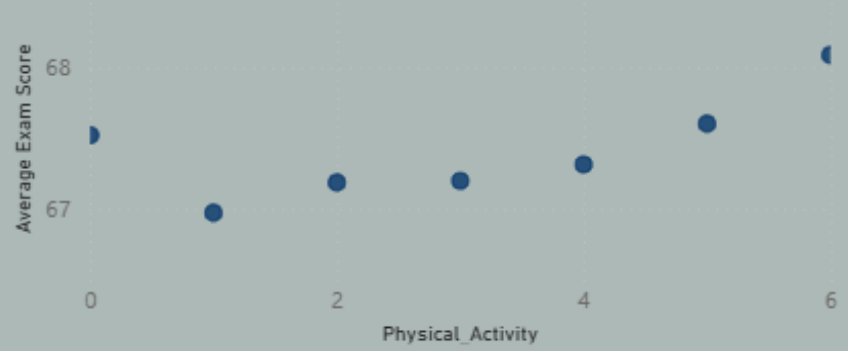
Extracurricular_A...: All

Distance_from_Home: All

7.03
Average Sleep Hours

2.97
Average of Physical_Activity

Average Exam Score by Physical_Activity



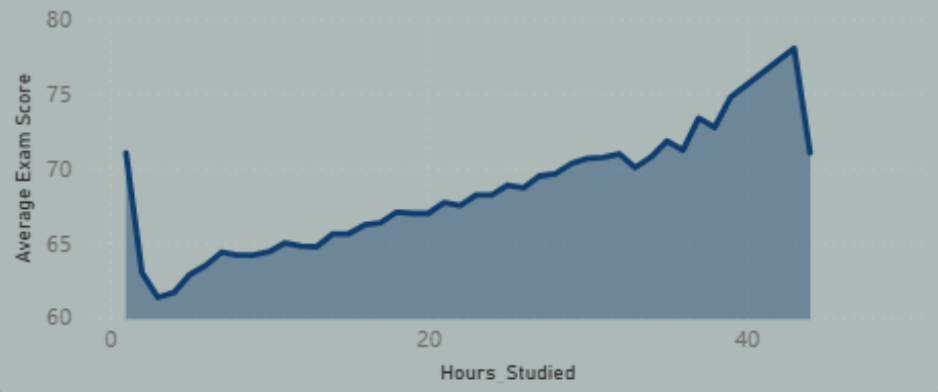
Average sleep hours by Performance and physical activity

Performance category	0	1	2	3	4	5	6
Average Achiever	7.07	6.93	7.04	7.06	7.00	7.01	6.95
High Achiever	7.19	7.17	6.95	7.05	7.02	6.93	8.00

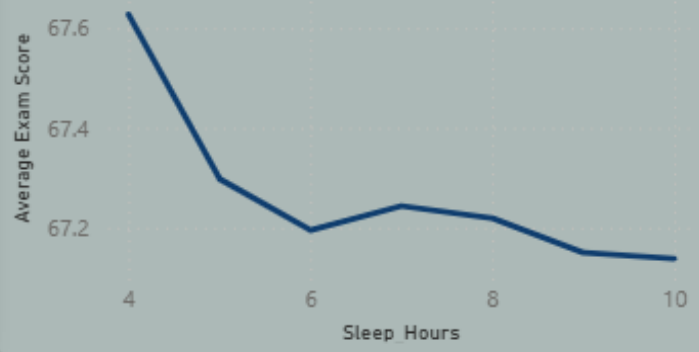
Performance by Extracurricular participation

Extracurricular_Activities	Total Students	Average of Sleep_Hours	Average Exam Score
No	2669	7.03	66.93
Yes	3938	7.03	67.44

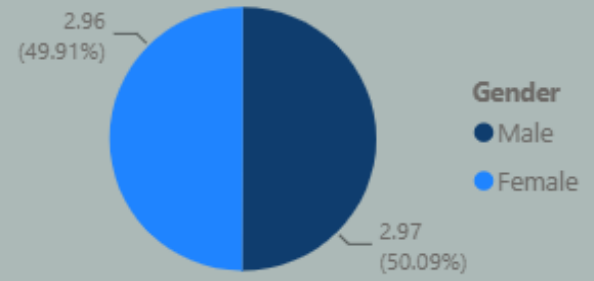
Average Exam Score by Hours_Studied



Average Exam Score by Sleep_Hours



Average of Physical_Activity by Gender



Executive Overview | School Environment & Student Success | Academic Performance | Family & Resource Analysis | Lifestyle & Wellbeing